CO 2 – ASSESSMENT TOOL -3

COURSE CODE: **DSA0308**

COURSE NAME: **NLP**

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Scenario-Based questions:

Q3. Error Analysis of Stemming in News Article Classification

1. How does stemming affect feature extraction?

Stemming reduces words to their root or stem form. It helps reduce the number of features in a dataset. However, incorrect stemming can remove important meanings and affect classification accuracy.

2. Identify incorrect stems generated by Porter Stemmer.

Original Word	Stem	Problem
organization	organ	Meaning is lost
organizer	organ	Meaning is lost
organizing	organ	Meaning is lost
organized	organ	Meaning is lost
organization's	Depends on tokenization	Possessive form

3. Would lemmatization improve classification?

Yes. Lemmatization uses vocabulary and linguistic rules to convert words into meaningful base forms.

Examples:

- organizing → organize
- organized → organize
- organizations → organization

Lemmatization preserves meaning better than stemming in many cases.

4. Compare preprocessing methods.

Method	Advantages	Disadvantages
Without stemming	Preserves original words	Large vocabulary
Porter stemming	Reduces vocabulary size	May lose meaning
Lemmatization	Produces meaningful base forms	Takes more processing time

5. Conclusion

Lemmatization is a good choice when preserving semantic meaning is important. The best method should be selected by comparing classification accuracy on the dataset.

Q4. Morphological Error Analysis in an Information Retrieval System

1. Analyze stemming errors affecting search relevance.

Stemming removes word endings to find related words. Incorrect stemming can produce irrelevant search results and reduce search accuracy.

2. Categorize each error.

Word	Root	Morphology
watches	watch	Inflectional
watching	watch	Inflectional
washable	wash	Derivational
washer	wash	Derivational
washed	wash	Inflectional

3. Why should certain suffixes not be removed?

Inflectional suffixes change grammatical forms, while derivational suffixes can change a word's meaning or grammatical category.

For example:

Watching: A form of the verb *watch*.

Washer: A device or person that washes.

Washable: Something that can be washed.

Removing all suffixes may combine words with different meanings, reducing product-search relevance.

4. Propose an improved morphological preprocessing strategy.

1. Convert all words to lowercase.
2. Remove unnecessary punctuation.
3. Tokenize the text into words.
4. Use lemmatization for appropriate inflectional forms.
5. Preserve important derivational distinctions.
6. Test the search system using precision and recall.

Conclusion: An improved morphological preprocessing strategy increases search relevance and helps users find the correct products.

Q5. Correct the Porter Stemmer Program

1. Identify the coding errors.

The original code is:

```
from nltk.stem import PorterStemmer
import pandas as pd

ps = PorterStemmer()

data = pd.read_csv("BBCNews.csv")

data["Processed"] = data["Text"].apply(ps.stem)

print(data["Processed"].head())
```

Errors:

1. ps.stem() expects a single word, not a complete news article.
2. The text is not tokenized into individual words.
3. Punctuation is not handled.
4. The program assumes the CSV file contains a column named Text.

2. Why is the output incorrect?

The stemmer is designed to process individual words. Applying it directly to an entire article does not correctly stem each word separately.

3. Corrected program

```
import re

from nltk.stem import PorterStemmer


ps = PorterStemmer()


text = "connected connection connecting connectivity studies studied studying study"


words = re.findall(r"[a-zA-Z]+", text.lower())


print("Original Word -> Stemmed Word")


for word in words:
    print(word, "->", ps.stem(word))
```

OUTPUT:

```
= RESTART: C:/Users/kakar/AppData/Local/Programs/Python/Python314/co2 at3 q1.py
Original Word -> Stemmed Word
connected -> connect
connection -> connect
connecting -> connect
connectivity -> connect
studies -> studi
studied -> studi
studying -> studi
study -> studi
```

Conclusion: The corrected program tokenizes the text before stemming, producing properly processed words. Lemmatization may be preferable when preserving meaning is important.

Q6. Error Analysis of a Finite-State Morphological Parser

1. Identify the logical errors.

Original code:

```
def parser(word):
    if word.endswith("s"):
        return word[:-2], "Plural Noun"
    else:
        return word, "Singular"

words = ["cars", "boxes", "cities", "children"]

for w in words:
    print(parser(w))
```

Error: word[:-2] removes two characters instead of one for regular plurals.

For example, cars becomes ca instead of car.

2. Corrected Python program

```
irregular = {
    "children": "child",
    "men": "man",
    "women": "woman",
    "mice": "mouse",
    "feet": "foot",
    "teeth": "tooth",
    "people": "person"
}

def parser(word):
    word = word.lower()

    if word in irregular:
        return word, irregular[word], "Plural Noun"
```

```
    elif word.endswith("ies"):
        return word, word[:-3] + "y", "Plural Noun"

    elif word.endswith(("ches", "shes", "xes", "zes", "sses")):
        return word, word[:-2], "Plural Noun"

    elif word.endswith("s") and not word.endswith("ss"):
        return word, word[:-1], "Plural Noun"

    else:
        return word, word, "Singular or Unknown"

words = ["cars", "boxes", "cities", "children"]

for w in words:
    print(parser(w))
```

OUTPUT:

```
= RESTART: C:/Users/kakar/AppData/Local/Programs/Python/Python314/co2 at3 q2.py
('cars', 'car', 'Plural Noun')
('boxes', 'box', 'Plural Noun')
('cities', 'city', 'Plural Noun')
('children', 'child', 'Plural Noun')
```

Conclusion: The corrected parser handles regular plurals, -es, -ies, and selected irregular plurals. A lexical database such as WordNet can improve its accuracy.